

Course Name: Digital electronics and Computer Organisation

Course Code: **Internal Marks: 25** **External Marks: 75** **Credit: 4**

UNIT-I

Logic gates and circuit:- Gates (OR, AND, NOR, NAND, XOR & XNOR); Demorgan's laws; Boolean laws, Circuit designing techniques (SOP, POS, K-Map).

UNIT-II

Combinational Building Blocks:- Multiplexes; Decoder; Encoder; Adder and Subtractor.

UNIT-III

Memories:- ROMs, PROMs, EPROMs, RAMs, Hard Disk, Floppy Disk and CD-ROM.

UNIT-IV

Sequential Building Blocks:- Flip-Flop (RS, D, JK, Master-slave & T flip-flops); Registers & Shift registers; Counters; Synchronous and Asynchronous Designing method.

UNIT-V

Memory Organization:- Basic cell of static and dynamic RAM; Building large memories using chips; Associative memory; Cache memory organisation and Virtual memory organisation.